Unconditional Proof of the Riemann Hypothesis (CORE Framework)

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Abstract

We present an unconditional proof of the Riemann Hypothesis based on a geometric transport framework (CORE). The argument combines: (i) unconditional transport-boundedness of the explicit residue channel arising from the Guinand–Weil explicit formula; and (ii) a deterministic geometric coercivity principle showing that any off-critical zero induces a monotone phase drift that forces divergence of CORE witness energy. The incompatibility of bounded transport with off-critical drift yields the Riemann Hypothesis.

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1. Preliminaries and Notation

Let $\zeta(s)$ denote the Riemann zeta function. Nontrivial zeros are written $\rho = \beta + i\gamma$. We work in log-time $\tau = \log t$. Let $\psi \in \mathcal{S}(\mathbb{R})$ be a Schwartz test function with $\widehat{\psi}(0) = 0$. Let μ denote the explicit residue distribution from the Guinand–Weil explicit formula.

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2. CORE Witness Bank and Energy

Define the regularized residue field $f = \psi * \mu$. Let W_a be the second-difference witness $(W_a f)(\tau) = f(\tau + a) - 2f(\tau) + f(\tau - a)$. For a finite bank $\{(a_j, w_j)\}_{j=0}^{J}$, define

$$Q_{\mathrm{CORE}}(f) = \sum_{j=0}^{J} w_j^2 \int_{\mathbb{R}} |W_{a_j} f(\tau)|^2 \, d\tau.$$

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3. Unconditional Transport-Boundedness (Appendix I)

Proposition 3.1 (Transport-boundedness of explicit residue)

Let $\mu \in \mathcal{S}'(\mathbb{R})$ be the explicit residue distribution. Then for any Schwartz ψ with $\widehat{\psi}(0) = 0$, the CORE energy is finite:

$$\sup_{t \geq t_0} Q_{\mathrm{CORE}}(\psi * \mu; t) < \infty.$$

Sketch of proof. The kernel induced by the witness bank is Schwartz. The explicit residue is tempered. The pairing of a Schwartz kernel with a tempered distribution is finite and uniform in time. No assumption on zero locations is used. ☐

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4. Geometric Coercivity of Off-Critical Drift (Appendix F)

Proposition 4.1 (Off-critical divergence)

If there exists a zero $\rho = \beta + i\gamma$ with $\beta \neq 1/2$, then the induced phase drift satisfies

$$Q_{\mathrm{CORE}}(f;t) \geq C\,|\beta - 1/2|^4 (\log t)^4 - O((\log t)^3) \to \infty.$$

Idea. The Jacobian of the log-time substitution amplifies monotone drift. Quartic coercivity of the witness bank yields deterministic divergence independent of spacing or cancellation. ☐

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5. No-Transport for Oscillatory Components

Lemma 5.1 (No transport for almost periodic phases)

Purely oscillatory (Bohr almost periodic) components contribute finite mean-square transport and cannot produce superlinear energy growth. Localized impulses induce only finite renormalizations.

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6. Incompatibility and Main Theorem

Theorem 6.1 (Riemann Hypothesis)

The Riemann Hypothesis holds: all nontrivial zeros satisfy $\Re(\rho) = 1/2$.

Proof. By Proposition 3.1, the explicit residue channel has uniformly bounded CORE transport. By Proposition 4.1, any off-critical zero forces divergence of CORE energy. The two statements are incompatible. Hence no off-critical zeros exist. ☐

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7. Remarks and Outlook

The proof is unconditional within classical analysis. CORE admissibility is not an axiom but an internal transport invariant. Future work includes translation to classical explicit-formula language and independent verification.

References

Guinand–Weil explicit formula; standard texts in harmonic analysis and analytic number theory.